Location Privacy Challenges and Solutions : Location Transformation, MAC Rotation, and other solutions

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November 2025, Stockholm Sweden.

***Abstract*—This research-based paper will investigate anonymity and privacy threats posed by location based services. This paper will also cover some solutions from user end to new protocols put in place and other services accessibility like third-parties. This paper aims to create a foundation for future research and help clarify how the different objectives can be accomplished.**

INTRODUCTION

The ubiquity of Location-Based Services (LBS), such as Google Maps, has revolutionized how individuals navigate, interact, and access information in real time. While these services offer unparalleled convenience and efficiency, they also introduce significant risks to user anonymity and privacy. The continuous collection, processing, and sharing of location data expose users to potential threats, including unauthorized surveillance, profiling, and malicious exploitation of personal information[1]. These concerns not only decrease user trust but also pose ethical and regulatory challenges for service providers.

This research-based paper examines the anonymity and privacy threats inherent in LBS, analyzing the technical and social implications of location data exposure. Additionally, this paper evaluates existing solutions-ranging from user-end strategies like anonymization techniques and privacy settings to intermediary services that aim to mitigate risks without compromising functionality. Our goal is to foster a more secure and trustworthy ecosystem for users while not compromising the efficiency of the service.

Problem Definition

This study aims to create a foundation for future research and help clarify how the different objectives can be accomplished.

Type of attacks possible: From stalking to house burglary through traffic manipulation [1], the exploitation of location data in LBS exposes users to a spectrum of threats that extend beyond digital privacy into real-world harm. The interconnected nature of modern LBS means that vulnerabilities in one application, such as weak encryption in a navigation app or lax privacy controls on a social platform, can cascade into systemic risks, affecting everything from personal security to public infrastructure.

State of the art: Current research in this area can help us understand how attackers operate and the methods they use. Studies from other related fields may also provide useful background information. Additionally, non-academic research (like industry reports or practical case studies) can offer real-world insights that support academic findings. Theses valuable findings will be collected and processed.

Focus on solution: Few solutions will be explained, some from the user side like MAC rotation or location granularity control [2] and also new possible ways to implement location in fully anonymity with new methods like cloud GNSS [3] and authentication during the location process or even P2P (peer to peer) location sharing. We'll have a deep focus on anonimizers entities and dummy location system [4]

This section explores a range of solutions designed to enhance privacy and anonymity in Location-Based Services (LBS). On the user side, techniques such as MAC address rotation and location granularity control empower individuals to limit their exposure to tracking. Additionally, we examine innovative approaches for achieving fully anonymous location services, including cloud-based GNSS processing, authentication mechanisms embedded within the location process, and peer-to-peer (P2P) location sharing. Particular emphasis will be placed on anonymizer entities—intermediaries that obscure user identities—and dummy location systems, which generate decoy data to further protect privacy.

Technical analysis: While some solutions may appear intuitive or straightforward at first glance, others demand a thorough examination to fully understand their implications. This section deal with the critical questions that arise when assessing such solutions. For instance, how does a particular method contribute to improving privacy or anonymity? Does it effectively obscure sensitive information, or does it merely provide a superficial layer of protection? Additionally, performance considerations are paramount: does the solution introduce latency or computational overhead, and if so, what is the tangible cost in terms of system efficiency or user experience? Finally some solutions may be theoretically possible but prove challenging to deploy in real-world scenarios due to technical constraints, compatibility issues, or resource limitations. By addressing these questions, we aim to provide a more tangible understanding of the strengths and limitations of each approach.

Expected Results

This study is expected to give a comprehensive overview of key ideas behind the urge to anonymize data from location based services, an analysis of already existing solutions via technical figures or explanation and finally this paper aim to help any start for a thesis.

This study aims to provide a clear understanding of why anonymizing location-based data matters in today's digital world. As apps and services increasingly rely on location data, the need to protect user privacy has never been more urgent. This paper will break down the existing solutions, explaining how they work and evaluating their effectiveness using real-world examples and technical insights. Our goal is to make this research useful for anyone interested in the topic, whether you're a student starting a thesis, a developer building privacy tools, or simply someone curious about data protection. By summarizing what we know and pointing out where more work is needed, we hope to encourage new ideas and help shape a future where location-based technologies respect user privacy.

Novelty

This study seeks to establish a foundation for future research and to provide clarity on how the various objectives in this field can be effectively achieved. Rather than introducing a groundbreaking new encryption method within a limited timeframe of three months, the novelty of this paper lies in its comprehensive synthesis of the current state of research on privacy in GNSS. By consolidating existing knowledge and insights, this work aims to make these complex concepts more accessible and understandable. Ultimately, it is intended to serve as a valuable resource and potential starting point for further academic exploration, such as a thesis.

RELATED WORK

Some research paper are out on internet and this subject is studied since at least 2010 and so now some protocols are already put in place. Through carefull research about the number of citation of those articles, some seems to be major paper that changed the vision of GNSS. Regarding user technique (like MAC rotation) few papers have written about that, however more papers talk about server-side technique like new protocols or encryption to protect user data. The review paper is valuable as it provides a comprehensive analysis of few solutions to anonymize data provided by Location Base services. Books on KTH university covers this topic , it will help us to understand from the calculation of the location to the encryption of signals in order to prevent men-in-the-middle attack.

A significant body of research on this topic has been available online since the 2010's, and several protocols have already been implemented as a result. By carefully analyzing the citation counts of these studies, it becomes evident that some papers have had a major impact on the privacy in GNSS. While user-focused techniques, such as MAC rotation—have received relatively little attention in the literature, there is a much broader discussion around server-side approaches, including new protocols and encryption methods designed to safeguard user data. This review paper holds[2] particular value as it offers a thorough and comprehensive analysis of the limited yet innovative solutions available for anonymizing data provided by Location-Based Services. Additionally, resources from KTH University delve into this subject, providing insights that range from the technical calculations involved in determining location to the encryption of signals aimed at preventing man-in-the-middle attacks.

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